

Engineering Change Order ECO-2025-091

Product: ACT-BP-4500 Drillable Bridge Plug (also applies to -5500 slips) | Revision C → D

ECO Number / Date	ECO-2025-091 / 2025-10-07
Originator	A. Gutierrez
Affected Product	ACT-BP-4500 Drillable Bridge Plug (also applies to -5500 slips)
Revision Change	C to D
Change Classification	Class 2 (reliability improvement)
Effectivity	Lots received after 2025-06-15 (supplier Fort Worth Foundry Co.).

1. Reason for Change

Two field reports (FR-25-031, FR-25-046) of plug slippage during 10,000 psi casing tests in P110 casing. Teardown found slip insert hardness 48-51 HRC (drawing then specified 50 HRC min). P110 casing hardness approaches uncontrolled insert hardness, reducing bite.

2. Description of Change

Upgrade slip inserts (ACT-BP-4500-03/-04 and BP-5500 equivalents) from 'induction hardened 50 HRC min' to 'AISI 52100 through-hardened inserts, 55 HRC minimum, individually pressed and staked'. Add insert hardness to lot acceptance criteria (sample 13 pcs/lot, AQL 1.0). Drawing D-BP-4500-03 Rev C -> D; D-BP-5500-03 Rev B -> C.

3. Affected Documents

Document	From Rev	To Rev	Change Summary
BOM-BP-4500	C	D	Items 4-5 material note
Dwg D-BP-4500-03/-04	C	D	Insert spec & hardness
Dwg D-BP-5500-03/-04	B	C	Insert spec & hardness
QCP-BP-01 (lot plan)	B	C	Add D4 hardness characteristic
DS-BP-SERIES	C	D	Materials table

4. Validation / Qualification Evidence

Slip bite verified in P110 coupon fixture at 12,000 psi differential equivalent load, 3 samples, zero displacement (Test report TR-25-166). First production lot inspected under QIR-2025-0198 (initial lot rejected for low hardness and reworked - see NCR-2025-077; re-inspection QIR-2025-0205 accepted).

5. Disposition of Existing Stock and WIP

Rev C stock: restrict to <= L80 casing per EB-25-07; rework by insert replacement at next redress.

6. Implementation Checklist

Task	Owner	Status
Update drawings and BOM in PLM	Design Engineering	COMPLETE
Update datasheet and manual	Document Control	COMPLETE

Task	Owner	Status
Update router / work instructions	Manufacturing Engineering	COMPLETE
Train assembly & QC personnel	Quality	COMPLETE
Notify affected customers / bulletin	Sales Engineering	COMPLETE
First-article inspection on next lot	Quality	SCHEDULED

7. Approvals

Role	Name	Signature	Date
Engineering	J. Martinez, P.E.	/signed electronically/	2025-10-06
Quality	S. Nguyen	/signed electronically/	2025-10-06
Manufacturing	T. Okafor	/signed electronically/	2025-10-07
Doc Control	L. Hargrove	/signed electronically/	2025-10-07